KUET · ESE 3105 · Heat and Mass Transfer

ESE 3105 Question Map

Every question from the 2019, 2020, 2021, 2022, 2023 term papers, grouped by topic and ranked by how often it comes back and how many marks it carries. Answers are in the ESE 3105 Study Pack (HTML and PDF) next to this file.

24-hour list | Pass plan | Frequency grid | Section A | Section B

24-hour priority list

You answer **3 of 4 questions in each part**, so each part has its own list. Study down both lists until the green **pass line**: about 3 h for Part A and 4 h for Part B, which leaves time for sleep and a last read of the one-shot notes. Alternate between parts (A block 1, B block 1, A block 2 …) so neither is left short. **Chance**: how often the topic came up, newer papers weigh more (2023 ×5, 2022 ×4, 2021 ×3, 2020 ×2, 2019 ×1). **Covered**: share of that part's past-paper marks you have studied once you finish the row.

Retake tip: read the Concept Primer first (30 minutes). In Part A the wall/pipe resistance numerical (A2), critical radius (A3) and the fin derivation plus numerical (A4) come every year and all have calculators. The boiling curve (A6) is pure theory: learn the five regions. In Part B learn the dimensionless numbers (B1) and the flat-plate numerical (B3); natural convection on a plate (B6) is the easiest numerical. Tube banks (B5) are long: do them last.

Section A: Conduction, insulation, fins, transient, boiling and condensation

Block 1: very likely and easy · 1.2 h

Topic	Chance	Level	Covered
A6 Boiling and condensation	Very likely 100%	Easy	26%
A2 Steady one-dimensional conduction: walls, cylinders, spheres, thermal resistance, heat generation	Very likely 100%	Easy	44%

A1 Conduction basics: modes, k and α, boundary conditions, general conduction equation	Very likely 87%	Easy	59%

Block 2: very likely, medium effort · 2.2 h

Topic	Chance	Level	Covered
A4 Fins (extended surfaces): equation, solutions, efficiency, effectiveness, numericals	Very likely 100%	Medium	74%
A3 Critical radius (thickness) of insulation	Very likely 100%	Medium	82%
✔ Pass line: 82% of Section A marks covered, enough to choose 3 of the 4 questions (about 2.8 h)			
A5 Transient conduction: lumped system, Heisler/one-term solution, semi-infinite solid	Very likely 87%	Medium	100%

Section B: Convection and mass transfer

Block 1: very likely and easy · 0.4 h

Topic	Chance	Level	Covered
B1 Convection fundamentals: mechanism, boundary layers, dimensionless numbers, film temperature	Very likely 100%	Easy	22%

Block 2: very likely, medium effort · 1.5 h

Topic	Chance	Level	Covered
B7 Mass transfer: Fick's law, diffusion, analogies, numericals	Very likely 100%	Medium	43%
B2 Boundary-layer analysis: integral momentum and energy equations, Nu from profiles, Reynolds-Colburn analogy	Very likely 80%	Medium	60%

Block 3: very likely numericals · 1.0 h

Topic	Chance	Level	Covered
B5 Flow across tube banks	Very likely 87%	Hard	73%

Block 4: likely · 1.5 h

Topic	Chance	Level	Covered
B6 Natural (free) convection	Likely 67%	Medium	81%
✔ Pass line: 81% of Section B marks covered, enough to choose 3 of the 4 questions (about 3.7 h)			
B4 Internal forced convection: fully developed flow, entry lengths, analogy, pipe numericals	Likely 60%	Medium	93%

Only if time is left · 0.8 h

Topic	Chance	Level	Covered

B3 External forced convection over a flat plate: numericals	Possible 47%	Medium	100%

Pass plan

Five papers (2019-2023). Normal papers: 210 marks, answer any THREE of four questions in each section (35 marks each). 2020 was a short exam (120 marks, any TWO of three per section, 30 marks each). Section A (Q1-Q4) = conduction (equation, walls/pipes, insulation, fins), transient conduction, boiling and condensation. Section B (Q5-Q8) = convection (boundary layers, flat plate, pipes, tube banks, natural convection) and mass transfer. **Minimum pass set:** Section A: Q1 (modes + conduction equation + wall/pipe numerical), Q2 (critical insulation + fins), Q4 (boiling curve + condensation + condenser numerical). Section B: Q5 (convection mechanism, boundary layers, flat-plate numerical), Q7 (tube bank numerical or natural-convection plate), Q8 (Fick's law + a diffusion numerical). The course notes (Mahmudul sir, Mahfuz notes, Sohag sir) follow Cengel's *Heat and Mass Transfer*, so answers use Cengel's methods and table values; every numerical was re-checked in Python.

SECTION A: CONDUCTION, INSULATION, FINS, TRANSIENT, BOILING AND CONDENSATION

5 topics

cover 82% of all Section A marks asked

- Q1: modes, k and α, derive the conduction equation, wall/pipe numerical
- Q2: critical radius derivation, fin derivation, fin numerical
- Q4: boiling curve, film vs drop-wise, condenser numerical
- Backup Q3: lumped derivation + shaft (one-term) numerical

SECTION B: CONVECTION AND MASS TRANSFER

5 topics

cover 71% of all Section B marks asked

- Q5: convection mechanism, boundary layers, h decreasing with x, flat-plate numerical
- Q7: tube bank numerical (Zukauskas) or Q6 natural convection on a plate
- Q8: Fick's law, absorption vs adsorption, diffusion numerical
- Backup: Nu/Pr/Gr significance, turbulent boundary layer

- **Section A, pick Q1, Q2, Q4:** Q1 = modes/k/α + derive the conduction equation + a wall or pipe resistance numerical (A1, A2); Q2 = critical radius + fin derivation + fin numerical (A3, A4); Q4 = boiling curve + film vs drop-wise + condenser numerical (A6). Q3 (transient, A5) is the backup: its lumped derivation and steel-shaft numerical repeat.
- **Section B, pick Q5, Q7 (or Q6), Q8:** Q5 = convection mechanism, boundary layers, flat-plate numerical (B1, B3); Q7 = tube bank numerical (B5) or Q6 = dimensionless numbers + natural-convection plate (B1, B6); Q8 = Fick's law, absorption vs adsorption, diffusion numerical (B7).
- **Repeated numericals:** brick wall 3×6×0.25 m (2022, 2023); stainless shaft 35 cm, 400 → 150 °C (2019, 2023); condenser 33 tubes, 12.35 kPa (2022, 2023); 0.6 m square plate natural convection (2019, 2023).
- **Repeated derivations:** conduction equation (cylindrical 2×, Cartesian 1×), critical radius (4×), fin heat loss/efficiency (2×), lumped $\theta/\theta_i = e^{-Bi \cdot Fo}$ (2×), boundary-layer energy equation (2×), fully developed axial gradient (2×).
- **Always draw:** the boiling curve, thermal-resistance network, fin sketch, boundary layer on a plate, tube bank layout.

Frequency grid

Marks asked from each topic in each paper. Darker means more marks. Click a topic to jump to its questions.

Marks per paper: 1–8 9–16 17–24 25+

Sort Section A: Best order (likely + easy) | Most likely first | Easiest first | Paper order

SECTION A: CONDUCTION, INSULATION, FINS, TRANSIENT, BOILING AND CONDENSATIO PAPER

Topic	2019	2020	2021	2022	2023	Years	Marks	
A6 Boiling and condensation	30	35	35	35	35	5/5	170	Pass s
A2 Steady one-dimensional conduction: walls, cylinders, spheres, thermal resistance, heat generation	29	15	17	20	36	5/5	117	Pass s

A1 Conduction basics: modes, k and α, boundary conditions, general conduction equation	29		18	27	22	4/5	96	Pass
A4 Fins (extended surfaces): equation, solutions, efficiency, effectiveness, numericals	15	30	20	8	20	5/5	93	Pass se
A3 Critical radius (thickness) of insulation	3	10	15	15	12	5/5	55	Pass se
A5 Transient conduction: lumped system, Heisler/one-term solution, semi-infinite solid	29		35	35	15	4/5	114	Next to a

Sort Section B: Best order (likely + easy) | Most likely first | Easiest first | Paper order

SECTION B: CONVECTION AND MASS TRANSFER: MARKS ASKED PER PAPER

Topic	2019	2020	2021	2022	2023	Years	Marks	
B1 Convection fundamentals: mechanism, boundary layers, dimensionless numbers, film temperature	23	18	33	39	30	5/5	143	Pass s
B7 Mass transfer: Fick's law, diffusion, analogies, numericals	35	22	35	8	35	5/5	135	Pass se
B2 Boundary-layer analysis: integral momentum and energy equations, Nu from profiles, Reynolds-Colburn analogy	15	30		39	25	4/5	109	Next to a
B5 Flow across tube banks	20		20	20	25	4/5	85	Pass
B6 Natural (free) convection	15			22	15	3/5	52	Pass
B4 Internal forced convection: fully developed flow, entry lengths, analogy, pipe numericals	17		52		10	3/5	79	Next to
B3 External forced convection over a flat plate: numericals	15	20		12		3/5	47	Pass s

Section A: Conduction, insulation, fins, transient, boiling and condensation

questions 1-4

Sort Section A: Best order (likely + easy) | Most likely first | Easiest first | Paper order

A6 Boiling and condensation

Pass set · Very likely · 100% · Easy

Asked in 5 of 5 papers · 170 marks in total

STUDY THESE

- Pool boiling curve: natural convection, nucleate (isolated bubbles, jets/columns), transition, film boiling; critical heat flux (burnout) point
- Film-wise vs drop-wise condensation (drop-wise h up to 10 times higher but hard to maintain)
- Nusselt film condensation: vertical plate, horizontal tube $h = 0.729[g\rho_l(\rho_l - \rho_v)h^*_{fg}k^3/\mu(T_{sat} - T_s)D]^{1/4}$; N tubes: $h/N^{1/4}$; $h^*_{fg} = h_{fg} + 0.68c_p\Delta T$
- Film boiling on a wire: $q_{film} + \frac{3}{4}q_{rad}$

> 2023 Q4(a) 10 marks
>
> Differentiate between film and dropwise condensation; describe film condensation on a vertical plate.

> 2023 Q4(b) 10 marks
>
> Describe pool boiling with the diagram, mentioning each stage.

> 2023 Q4(c) 15 marks
>
> Same condenser (33 tubes per column, 1.5 cm, 12.35 kPa, 20 °C).

> 2022 Q4(b) 7 marks
>
> For condensation over a vertical plate, which mode is preferable and why? Explain with equations.

> 2022 Q4(a) 10 marks
>
> Explain briefly the physical mechanism of boiling with a diagram.

> 2022 Q4(c) 18 marks
>
> Condenser columns of 33 tubes, D = 1.5 cm, 12.35 kPa, tube surfaces 20 °C. Find (i) average h, (ii) condensation rate per unit length of a column.

2021 Q4(a) 8 marks

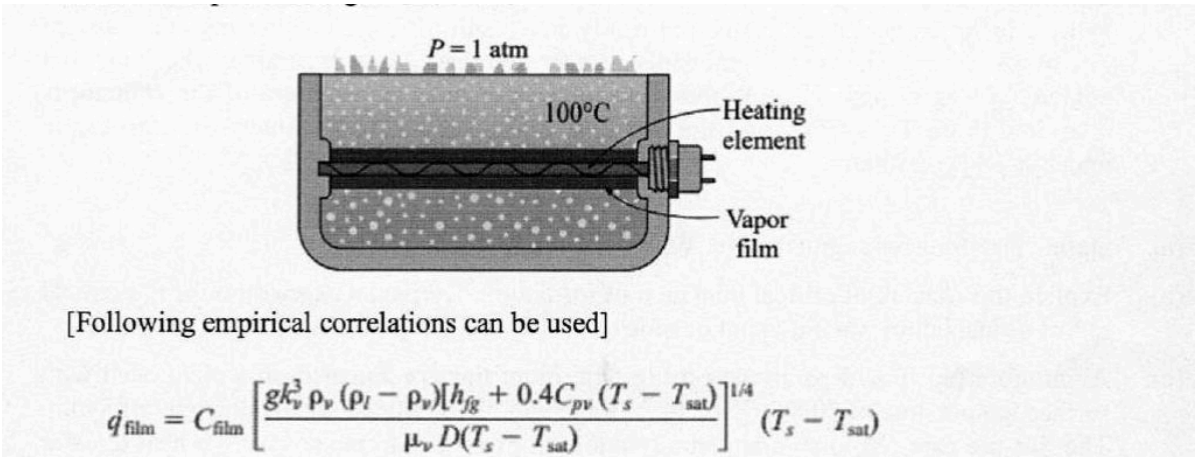
Distinguish between film-wise and drop-wise condensation. Which is more effective?

2021 Q4(b) 15 marks

Draw the boiling curve, identify regimes and explain each.

2021 Q4(c) 12 marks

Horizontal polished copper element D = 5 mm, ε = 0.05, at 350 °C in water at 1 atm: heat transfer per unit length (film boiling).



[Following empirical correlations can be used]

$$\dot{q}_{film} = C_{film}\left[\frac{g k_v^3 \rho_v (\rho_l - \rho_v)[h_{fg} + 0.4 C_{pv}(T_s - T_{sat})]}{\mu_v D(T_s - T_{sat})}\right]^{1/4}(T_s - T_{sat})$$

2020 Q2(b) 5 marks

Explain why condenser tubes are usually horizontal.

2020 Q3(b) 15 marks

Draw the boiling curve, identify the regimes and the burnout point. How is burnout caused and why is it avoided in boilers?

2020 Q3(a) 15 marks

Saturated steam at 85 °C condenses on 225 horizontal tubes (15 × 15), D = 1.27 cm, surface 75 °C; properties at 80 °C given. Condensation rate per m.

2019 Q4(a) 8 marks

Distinguish between (i) film-wise and drop-wise condensation, (ii) boiling and evaporation.

2019 Q4(b) 7 marks

Draw the pool boiling curve and show its regions.

2019 Q4(c) 15 marks

Condenser at 4.25 kPa, 100 horizontal tubes (10 × 10), 8 m long, D = 3 cm, surfaces at 20 °C. Find (i) heat transfer rate, (ii) condensation rate.

Full answer: ESE 3105 Study Pack, topic A6.

A2 Steady one-dimensional conduction: walls, cylinders, spheres, thermal resistance, heat generation

Pass set · Very likely · 100% · Easy

Asked in 5 of 5 papers · 117 marks in total

STUDY THESE

- Plane wall: $Q = kA(T_1 - T_2)/L$, linear $T(x)$; resistances $R_{wall} = L/kA$, $R_{conv} = 1/hA$ in series
- Cylinder $R = \ln(r_2/r_1)/(2\pi Lk)$; sphere $R = (r_2 - r_1)/(4\pi k\, r_1 r_2)$; overall coefficient $U = 1/(AR_{total})$
- Variable conductivity: use a mean k evaluated at the mean temperature (for linear k)
- Heat generation in a wire: $T_0 - T_s = \dot{e}r_0^2/4k$

> 2023 Q2(c) 11 marks
>
> Same brick wall (3 m × 6 m × 0.25 m, k = 0.8, 14 °C and 5 °C): rate of heat loss.

> 2023 Q1(c) 13 marks
>
> Plane wall with $k = k_0 t + 0.5k_0 t^2$ and faces at t_1, t_2: (i) derive the steady heat flow, (ii) temperature at which the mean k should be evaluated.

> 2023 Q2(a) 12 marks
>
> What is the overall heat transfer coefficient? Deduce its expression.

> 2022 Q1(c) 13 marks
>
> Plane wall L = 0.4 m, k = 1.8 W/m K, A = 30 m^2; left face 90 °C, right face loses heat by convection to 25 °C air with h = 24 W/m^2K. Find the heat transfer rate.

> 2022 Q2(d) 7 marks
>
> Brick wall 3 m × 6 m × 0.25 m, k = 0.8 W/m K, surfaces at 14 °C and 5 °C. Find the heat loss.

> 2021 Q1(c) 17 marks
>
> Resistance wire r = 0.2 cm, k = 15 W/m K, $\dot{e}$ = 50 W/cm^3, inside 0.5 cm ceramic (k = 1.2); outer surface 45 °C. Find centreline and interface temperatures.

> 2020 Q1(a) 15 marks

> Hollow sphere with surfaces at T_1 (r_1) and T_2 (r_2), constant k: develop the radial heat flow rate.

> 2019 Q1(c) 14 marks
>
> Plane wall L = 0.2 m, k = 1.2 W/m K, A = 15 m^2, T_1 = 120 °C, T_2 = 50 °C. Find (i) T at x = 0.1 m, (ii) heat conduction rate.

> 2019 Q2(c) 15 marks
>
> Steam at 320 °C in a cast iron pipe (k = 80), D_1 = 5 cm, D_2 = 5.5 cm, 3 cm glass wool (k = 0.05); h_i = 60, h_o = 18 W/m^2K, surroundings 5 °C. Find heat loss per m and temperature drops across pipe and insulation.

Full answer: ESE 3105 Study Pack, topic A2.

A1 Conduction basics: modes, k and α, boundary conditions, general conduction equation

Pass set · Very likely · 87% · Easy

Asked in 4 of 5 papers · 96 marks in total

STUDY THESE

- Modes: conduction (Fourier q = −k dT/dx), convection (Newton q = hΔT), radiation (Stefan-Boltzmann q = εσT^4)
- Thermal conductivity k (how well a material conducts) and thermal diffusivity α = k/ρc_p (how fast heat diffuses); steady vs transient
- Energy balance on a small element ⇒ general conduction equation in Cartesian, cylindrical, spherical coordinates; special forms (Fourier, Poisson, Laplace)
- Boundary conditions: specified temperature, specified flux, insulated, convection, radiation, interface; heat generation does not violate the first law; thermal contact resistance

> 2023 Q1(a) 8 marks
>
> What are the basic modes of heat transfer? Describe the modes by mentioning their governing equations.

> 2023 Q1(b) 14 marks
>
> Derive the general heat conduction equation for a rectangular coordinate system.

> 2022 Q1(a) 8 marks

> What are the basic modes of heat transfer? Write down the basic differences among them.

> 2022 Q1(b) 14 marks
>
> Derive the general heat conduction equation in cylindrical coordinates.

> 2022 Q2(a) 5 marks
>
> What is thermal contact resistance? Name the factors that affect it. (Critical thickness part in A3.)

> 2021 Q1(a) 10 marks
>
> Explain various kinds of boundary conditions commonly considered in heat conduction problems.

> 2021 Q1(b) 8 marks
>
> Does heat generation in a solid violate the first law of thermodynamics? Explain.

> 2019 Q1(a) 6 marks
>
> What are thermal conductivity and thermal diffusivity?

> 2019 Q2(a) 5 marks
>
> Distinguish between steady and transient heat transfer.

> 2019 Q1(b) 15 marks
>
> Derive the general heat conduction equation for a cylindrical coordinate system.

> 2019 Q3(a) 3 marks
>
> Define and explain thermal contact resistance (the critical-thickness part is in A3).

Full answer: ESE 3105 Study Pack, topic A1.

A4 Fins (extended surfaces): equation, solutions, efficiency, effectiveness, numericals

Pass set · Very likely · 100% · Medium

Asked in 5 of 5 papers · 93 marks in total

STUDY THESE

- Fin equation $d^2\theta/dx^2 - m^2\theta = 0$, $m = \sqrt{(hp/kA_c)}$, $\theta = T - T_\infty$
- Infinite fin $Q = \sqrt{(hpkA_c)}\theta_b$; insulated tip $Q = \sqrt{(hpkA_c)}\theta_b \tanh(mL)$; convective tip: use $L_c = L + A_c/p$
- Efficiency $\eta = Q_{fin}/Q_{fin,max} = \tanh(mL)/mL$; effectiveness $\varepsilon = Q_{fin}/(hA_b\theta_b)$

- Heat transfer augmentation; longer fin: more effective but less efficient

2023 Q3(a) 12 marks

What are fin effectiveness and efficiency? “Increasing fin length increases effectiveness but decreases efficiency”: justify.

2023 Q3(b) 8 marks

Motor body 400 mm OD, 250 mm long, surface ≤ 62 °C when dissipating 350 W; longitudinal fins 18 mm thick, 38 mm high; h = 42, air 35 °C, k = 40. Find the number of fins.

2022 Q2(c) 8 marks

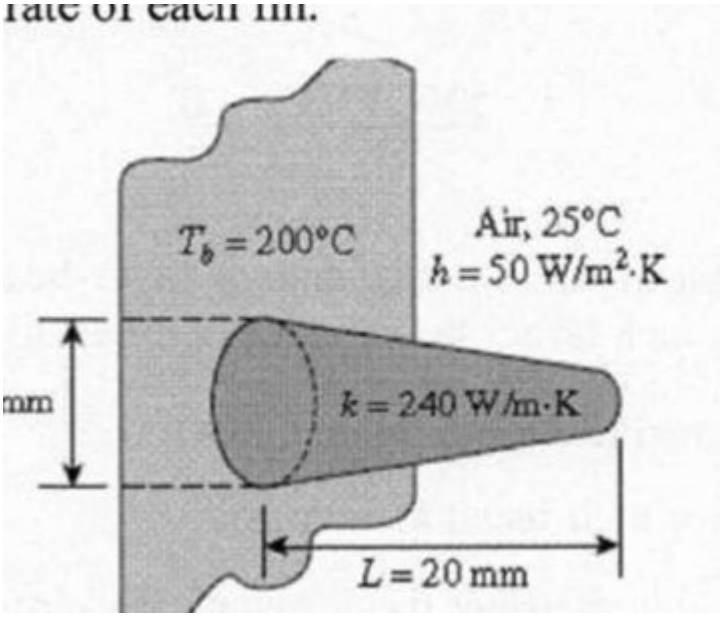
Explain (i) efficiency of fin, (ii) effectiveness of fin.

2021 Q2(a) 5 marks

Define heat transfer augmentation. Why is it necessary?

2021 Q2(c) 15 marks

Aluminium parabolic pin fins (blunt tip), L = 20 mm, D = 5 mm, k = 240, base 200 °C, air 25 °C, h = 50. Find efficiency and heat transfer per fin.



2020 Q2(c) 15 marks

Cylindrical fin, length L, insulated tip: derive the fin efficiency.

2020 Q1(b) 15 marks

Steel rod D = 2 cm, L = 25 cm, k = 50 W/m °C, h = 64 W/m^2°C, air 20 °C, base 120 °C. Find the heat loss.

2019 Q2(b) 15 marks

> Derive the heat loss from a fin of uniform cross-section with negligible tip heat loss.

Full answer: ESE 3105 Study Pack, topic A4.

A3 Critical radius (thickness) of insulation

Pass set · Very likely · 100% · Medium

Asked in 5 of 5 papers · 55 marks in total

STUDY THESE

- Adding insulation to a small pipe raises conduction resistance but also raises the outer area (lower convection resistance)
- Q is maximum at $r_{cr} = k/h$ (cylinder) and $2k/h$ (sphere)
- If $r_{outer} < r_{cr}$, removing the insulation *reduces* heat loss; electric wires use this to dissipate more heat

> 2023 Q2(b) 12 marks
>
> Significance of critical thickness of insulation; derive it for a cylindrical object.

> 2022 Q2(b) 10 marks
>
> Same question: insulation outer radius below critical, insulation removed; justify mathematically.

> 2022 Q2(a) 5 marks
>
> Define critical thickness of insulation.

> 2021 Q2(b) 15 marks
>
> Explain the concept of critical thickness of insulation. Derive the critical radius for a small hollow pipe.

> 2020 Q2(a) 10 marks
>
> Define critical radius of insulation. If the outer radius of insulation is less than critical and the insulation is removed, will heat transfer increase or decrease? Justify.

> 2019 Q3(a) 3 marks
>
> Define and explain critical thickness of insulation.

Full answer: ESE 3105 Study Pack, topic A3.

A5 Transient conduction: lumped system, Heisler/one-term solution, semi-infinite solid

Next to add · Very likely · 87% · Medium

Asked in 4 of 5 papers · 114 marks in total

STUDY THESE

- Lumped system valid if $Bi = hL_c/k < 0.1$ ($L_c = V/A_s$); $\theta/\theta_i = e^{-bt} = e^{-Bi\cdot Fo}$
- Large wall / long cylinder / sphere: one-term solution $\theta_0 = A_1 e^{-\lambda_1^2 \tau}$ (Heisler charts), heat transferred Q/Q_{max}
- Semi-infinite solid: $(T - T_s)/(T_i - T_s) = \text{erf}(\eta)$, $\eta = x/(2\sqrt{\alpha t})$; contact temperature of two solids

> 2023 Q3(c) 15 marks
>
> Same shaft problem (35 cm, 400 °C → 150 °C surroundings, h = 60, 20 min).

> 2022 Q3(a) 8 marks
>
> What is a lumped system? Copper ball D = 12 cm, k = 401, h = 15: can it be treated as lumped?

> 2022 Q3(b) 14 marks
>
> For a lumped system show $\theta/\theta_i = e^{-Bi\cdot Fo}$.

> 2022 Q3(c) 13 marks
>
> Brass plates 3 cm thick (k = 110, ρ = 8530, c_p = 380, $\alpha = 33.9\times10^{-6}$) at 25 °C, oven 700 °C for 10 min, h = 80. Find surface temperature.

> 2021 Q3(a) 17 marks
>
> Semi-infinite solid at T_i, surface suddenly at T_s: obtain $(T - T_s)/(T_i - T_s) = 1 - \text{erfc}(\eta)$ using the similarity variable.

> 2021 Q3(b) 10 marks
>
> Two semi-infinite solids A and B brought into contact: show $T_s = (T_{A,i} + \alpha T_{B,i})/(1 + \alpha)$, $\alpha = \sqrt{(k\rho c_p)_B}/\sqrt{(k\rho c_p)_A}$.

> 2021 Q3(c) 8 marks
>
> Snowpack at −10 °C for 60 days, soil k = 0.4, $\alpha = 0.15\times10^{-6}$, initially 15 °C: minimum burial depth of water pipes.

> 2019 Q3(b) 15 marks

For a lumped system show $\theta/\theta_i = e^{-Bi \cdot Fo}$.

2019 Q3(c) 14 marks

Stainless steel shaft D = 35 cm (k = 14.9, ρ = 7900, c_p = 477, α = 3.95×10^{-6}) from 400 °C cooled in 150 °C with h = 60. Centre temperature after 20 min and heat transfer per m.

Full answer: ESE 3105 Study Pack, topic A5.

Section B: Convection and mass transfer questions 5-8

Sort Section B: Best order (likely + easy) | Most likely first | Easiest first | Paper order

B1 Convection fundamentals: mechanism, boundary layers, dimensionless numbers, film temperature

Pass set · Very likely · 100% · Easy

Asked in 5 of 5 papers · 143 marks in total

STUDY THESE

- Convection = conduction + fluid motion; Newton's law $q = h(T_s - T_\infty)$; $h = -k(\partial T/\partial y)_{y=0}/(T_s - T_\infty)$
- Velocity and thermal boundary layers; laminar, transition, turbulent; developing vs fully developed flow; why h falls as δ_t grows
- Re (inertia/viscous), Pr (ν/α), Nu (convection/conduction), Gr (buoyancy/viscous), Ra = Gr·Pr; Nu = f(Re, Pr) by dimensional analysis
- Turbulent boundary layer: eddies mix fluid, steep wall gradient, h much larger; film temperature $T_f = (T_s + T_\infty)/2$; methods of finding h

2023 Q5(a) 10 marks

With a sketch, write down the physical mechanism of convection heat transfer.

2023 Q5(b) 12 marks

Differentiate hydrodynamic and thermal boundary layers; justify that h decreases as δ_t grows.

2023 Q6(a) 8 marks

Significance of (i) Prandtl, (ii) Rayleigh, (iii) Nusselt numbers.

2022 Q5(a) 11 marks

Explain the mechanism of convection. What are (i) hydrodynamic and (ii) thermal boundary layers?

2022 Q5(b) 12 marks

The thermal boundary layer thickens in the flow direction but h decreases. Why?

2022 Q6(b) 6 marks

Define film temperature and its significance.

2022 Q8(b) 10 marks

Methods of evaluating the convective coefficient; the analogy between heat and momentum transfer and the numerical method.

2021 Q5(a) 13 marks

Explain turbulent boundary-layer heat transfer. Why is the turbulent h much larger than laminar?

2021 Q6(b) 4 marks

Significance of Grashof number.

2021 Q6(a) 16 marks

Using dimensional analysis show Nu = f(Re, Pr) for forced convection over a flat plate.

2020 Q4(a) 10 marks

Physical significance of Nusselt and Prandtl numbers.

2020 Q6(b) 8 marks

Physical significance of Grashof number; how does it differ from Reynolds number?

2019 Q5(a) 5 marks

Distinguish between (i) thermal and hydrodynamic boundary layer, (ii) developing and developed flow.

2019 Q6(b) 12 marks

Explain turbulent boundary-layer heat transfer. Why is the turbulent h much larger than laminar?

2019 Q6(a) 6 marks

Significance of Nusselt and Prandtl numbers.

Full answer: ESE 3105 Study Pack, topic B1.

B7 Mass transfer: Fick's law, diffusion, analogies, numericals

Pass set · Very likely · 100% · Medium

Asked in 5 of 5 papers · 135 marks in total

STUDY THESE

- Fick's law $j_A = -\rho D_{AB} dw_A/dx$ (molar $J_A = -CD_{AB} dy_A/dx$); $D_{AB} \propto T^{3/2}/P$ for gases
- Mass transfer is driven by concentration difference (not bulk flow); heat-mass analogy (Fourier ↔ Fick, T ↔ C, α ↔ D_{AB}, Pr ↔ Sc, Nu ↔ Sh)
- Absorption vs adsorption; distillation
- Steady diffusion through a wall or spherical shell; equimolar counter-diffusion; diffusion of vapour through stagnant gas (Stefan tube): $N_A = (CD_{AB}/L)\ln[(1 - y_{A,L})/(1 - y_{A,0})]$

> 2023 Q7(a) 10 marks
>
> Analogy between heat and mass transfer; examples of (i) liquid-to-gas, (ii) solid-to-gas mass transfer.

> 2023 Q8(a) 8 marks
>
> State Fick's law; differences between absorption and adsorption.

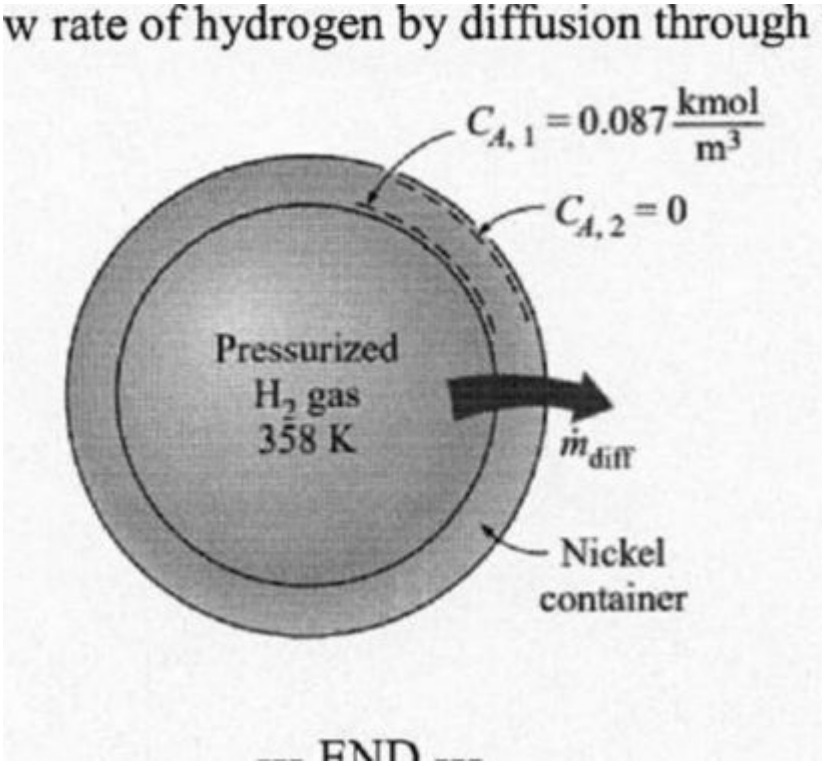
> 2023 Q8(b) 17 marks
>
> Toluene at the bottom of a 1.524 m deep vessel, air at 1 atm, 18.7 °C, D = 0.826×10^{-5} m^2/s, p_{sat} = 0.026 atm. Evaporation rate per unit area.

> 2022 Q8(a) 8 marks
>
> State Fick's law; differences between absorption and adsorption.

> 2021 Q8(b) 10 marks
>
> State Fick's law. How does gas mass diffusivity change with temperature and pressure?

> 2021 Q8(a) 10 marks
>
> Differences between absorption and adsorption; schematic of simple distillation.

> 2021 Q8(c) 15 marks
>
> H_2 at 358 K in a nickel sphere 4.8 m OD, 6 cm thick; C_A inside 0.087 $kmol/m^3$, outside 0. Mass flow rate of H_2 by diffusion.



2020 Q6(a) 12 marks

Define mass diffusivity. Explain Fick's law. Discuss analogies between heat and mass transfer.

2020 Q6(c) 10 marks

Difference between mass transfer and bulk fluid flow? Can mass transfer occur in a homogeneous medium?

2019 Q8(a) 6 marks

State and explain Fick's law of diffusion.

2019 Q8(b) 14 marks

Derive the diffusion rate for steady equimolar counter-diffusion of A and B.

2019 Q8(c) 15 marks

Water diffusing from the bottom of a beaker 10 cm diameter, 20 cm long, into dry air at 32 °C; D = 0.256 cm^2/s. Diffusion rate.

Full answer: ESE 3105 Study Pack, topic B7.

B2 Boundary-layer analysis: integral momentum and energy equations, Nu from profiles, Reynolds-Colburn analogy

Next to add · Very likely · 80% · Medium

Asked in 4 of 5 papers · 109 marks in total

STUDY THESE

- Cubic velocity profile $u/U = 1.5(y/\delta) - 0.5(y/\delta)^3 \Rightarrow \delta = 4.64x/\sqrt{Re_x}$; $C_{f,x} = 0.664/\sqrt{Re_x}$, average $1.328/\sqrt{Re_L}$
- Energy equation of the boundary layer: $u\partial T/\partial x + v\partial T/\partial y = \alpha\partial^2 T/\partial y^2$ (+ viscous dissipation); integral energy equation $d/dx\int u(T_\infty - T)dy = \alpha(\partial T/\partial y)_0$
- $\delta_t/\delta = Pr^{-1/3}/1.026$; $Nu_x = 0.332 Re_x^{1/2} Pr^{1/3}$, average = 2× local at L
- Reynolds-Colburn analogy $St \cdot Pr^{2/3} = C_f/2$; h from a given profile: $h = -k(\partial T/\partial y)_0/\Delta T$

> 2023 Q6(c) 12 marks
>
> Derive the energy equation for the laminar boundary layer on a flat plate.

> 2023 Q5(c) 13 marks
>
> Air at 20 °C over a plate at 160 °C with $(T - T_\infty)/(T_s - T_\infty) = e^{-ay}$, $a = 3200\ m^{-1}$. Find the surface heat flux and h.

> 2022 Q6(a) 14 marks
>
> Derive the integral energy equation for the thermal boundary layer and its thickness.

> 2022 Q6(c) 15 marks
>
> Rough plate with $h_x = ax^{-0.1}$: (i) ratio of average to local h at x, (ii) plot h_x and $\bar{h}_x$.

> 2022 Q7(a) 10 marks
>
> Concept and application of the Reynolds-Colburn analogy; its expression.

> 2020 Q4(b) 20 marks
>
> Laminar flow over a flat plate with the stated boundary conditions: determine the velocity boundary-layer thickness.

> 2020 Q5(a) 10 marks
>
> For laminar flow over a flat plate, develop the average friction coefficient.

> 2019 Q5(b) 15 marks
>
> With $\theta = 1.5(y/\delta_t) - 0.5(y/\delta_t)^3$ and $\delta_t = 4.53x/(Re_x^{1/2} Pr^{1/3})$, develop local Nu and average Nu from 0 to L.

Full answer: ESE 3105 Study Pack, topic B2.

B5 Flow across tube banks

Pass set · Very likely · 87% · Hard

Asked in 4 of 5 papers · 85 marks in total

STUDY THESE

- In-line or staggered; $V_{max} = S_T V/(S_T - D)$ (staggered: use $S_T V/2(S_D - D)$ if $S_D < (S_T + D)/2$)
- Zukauskas: $Nu = C\,Re^m Pr^n (Pr/Pr_s)^{0.25}$; row correction for $N_L < 16$; properties at mean temperature
- $T_e = T_s - (T_s - T_i)\exp(-A_s h/\dot{m}c_p)$; $Q = hA_s\Delta T_{lm}$; $\Delta P = N_L f\chi\rho V_{max}^2/2$

> **2023 Q7(b) 25 marks**
>
> Geothermal water at 120 °C in tubes; air 20 °C, 1 atm, 4.5 m/s; D = 1.5 cm in-line $S_L = S_T$ = 5 cm, 6 rows of 10 tubes. Heat transfer per unit length and pressure drop.

> **2022 Q7(c) 20 marks**
>
> Water 24 → 74 °C across a staggered bank, D = 2.5 cm, S_L/D = 1.5, S_T/D = 2, tubes at 100 °C, u_∞ = 0.3 m/s. Find (i) h, (ii) number of rows.

> **2021 Q7(b) 20 marks**
>
> Air 325 K, 8 m/s across in-line bank, D = 1.9 cm, $S_L/D = S_T/D$ = 2, tubes at 375 K, 15 rows, 20 tubes per row, 0.75 m. Find (i) pressure drop, (ii) exit temperature, (iii) heat transfer.

> **2019 Q7(a) 20 marks**
>
> Staggered bank D = 16.4 mm, S_L = 34.3 mm, S_T = 31.3 mm, 7 rows of 8 tubes, surface 70 °C, air 15 °C at 6 m/s, C_2 = 0.95, m = 0.6. Find h and heat transfer.

Full answer: ESE 3105 Study Pack, topic B5.

B6 Natural (free) convection

Pass set · Likely · 67% · Medium

Asked in 3 of 5 papers · 52 marks in total

STUDY THESE

- Buoyancy-driven flow; $Gr = g\beta\Delta TL^3/\nu^2$, $Ra = Gr \cdot Pr$; $\beta = 1/T_f$ for ideal gas
- Vertical plate (L = height): $Nu = 0.59Ra^{1/4}$ (10^4-10^9), $0.1Ra^{1/3}$ (turbulent), Churchill-Chu
- Horizontal plate ($L_c = A/p$): hot side up $0.54Ra^{1/4}$ / $0.15Ra^{1/3}$; hot side down $0.27Ra^{1/4}$

- Horizontal cylinder: $Nu = \{0.6 + 0.387Ra^{1/6}/[1 + (0.559/Pr)^{9/16}]^{8/27}\}^2$; forced vs natural convection

> 2023 Q6(b) 15 marks
>
> Same 0.6 m × 0.6 m plate problem (vertical, hot up, hot down).

> 2022 Q7(b) 5 marks
>
> Differences between forced and natural convection with examples.

> 2022 Q8(c) 17 marks
>
> 6 m long, 8 cm horizontal hot water pipe at 70 °C in a 20 °C room: natural convection heat loss.

> 2019 Q7(b) 15 marks
>
> 0.6 m × 0.6 m plate at 90 °C in a 30 °C room, other side insulated: heat transfer if (i) vertical, (ii) horizontal hot face up, (iii) hot face down.

Full answer: ESE 3105 Study Pack, topic B6.

B4 Internal forced convection: fully developed flow, entry lengths, analogy, pipe numericals

Next to add · Likely · 60% · Medium

Asked in 3 of 5 papers · 79 marks in total

STUDY THESE

- Hydrodynamic entry $L_h \approx 0.05ReD$, thermal $L_t \approx 0.05RePrD$ (laminar); $Pr > 1 \Rightarrow$ thermal entry longer
- Thermally fully developed: $\partial/\partial x[(T_s - T)/(T_s - T_m)] = 0$; with uniform q_s, $\partial T/\partial x = dT_m/dx = const$
- Turbulent: Dittus-Boelter $Nu = 0.023Re^{0.8}Pr^n$; roughness raises f only in turbulent flow; $St = f/8$ (Reynolds analogy)
- $T_e = T_s - (T_s - T_i)\exp(-hA_s/\dot{m}c_p)$; $\Delta P = f(L/D)\rho V^2/2$; $f = 64/Re$ laminar

> 2023 Q8(c) 10 marks
>
> “For thermally fully developed flow the axial temperature gradient is independent of the radial direction”: justify.

> 2021 Q5(b) 12 marks
>
> For forced convection inside a tube, prove $St = f/8$.

2021 Q5(c) 5 marks

How does surface roughness affect pressure drop in turbulent tube flow? And in laminar flow?

2021 Q5(d) 5 marks

Sketch hydrodynamic and thermal entry lengths for Pr = 0.7 and Pr = 7.

2021 Q7(a) 15 marks

Prove that the axial temperature gradient does not depend on radial position for uniform surface heat flux in thermally fully developed flow.

2021 Q6(c) 15 marks

Annulus 35 mm ID, 50 mm OD; water 20 → 34 °C at 2 m/s, inner surface 60 °C. Estimate h ($Nu = 0.023Re^{0.8}Pr^{n}$).

2019 Q6(c) 17 marks

Oil in a 30 cm pipeline at 2 m/s, 200 m through icy lake (surface 0 °C): (i) exit temperature, (ii) heat transfer rate, (iii) pumping power.

Full answer: ESE 3105 Study Pack, topic B4.

B3 External forced convection over a flat plate: numericals

Pass set · Possible · 47% · Medium

Asked in 3 of 5 papers · 47 marks in total

STUDY THESE

- Properties at the film temperature; $Re_L = VL/\nu$ laminar if $Re < 5\times10^5$
- Laminar average $Nu = 0.664Re_L^{1/2}Pr^{1/3}$; turbulent $Nu = 0.037Re^{0.8}Pr^{1/3}$; mixed $(0.037Re^{0.8} - 871)Pr^{1/3}$
- Low-pressure gas: density (and ν) changes with pressure; k, μ, Pr do not
- Unheated starting length: Nu_x multiplied by $[1 - (\xi/x)^{3/4}]^{-1/3}$

2022 Q5(c) 12 marks

Air at 5 m/s, 1 atm, 28 °C over a plate at 95 °C heated from x = 12 cm: total heat transfer up to x = 40 cm.

2020 Q5(b) 20 marks

Plastic sheet 1.2 m wide, 2 mm thick, 15 m/min, 90 °C, air 3 m/s at 30 °C on both sides, 2 s in the cooling section, ε = 0.9. Find the heat transfer rate.

2019 Q5(c) 15 marks

> Air at 6 kN/m^2, 300 °C, 10 m/s over a 0.5 m plate kept at 27 °C: cooling rate per unit width.

Full answer: ESE 3105 Study Pack, topic B3.

Abbreviations used in this pack

Every short form, written out. The answers also spell each one out the first time it appears.

Short form	Full meaning
BC	boundary condition
Bi	Biot number
CHF	critical heat flux
Fo	Fourier number
Gr	Grashof number
ID	inner diameter
kmol	kilomole
kPa	kilopascal
kW	kilowatt
LMTD	log mean temperature difference
Nu	Nusselt number
OD	outer diameter
Pr	Prandtl number
Ra	Rayleigh number
Re	Reynolds number
Sc	Schmidt number
Sh	Sherwood number
St	Stanton number

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